

Group 9 Assignment Report

An Evolutionary Algorithm for the Travelling Salesman Problem

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1 Introduction

The goal of this assignment is to design, implement and experimentally evaluate an evolutionary algorithm for a combinatorial optimisation problem. We chose the Travelling Salesman Problem (TSP), a classical NP-hard problem that consists of finding the shortest tour visiting every city exactly once. Its discrete, permutation-based search space and its relevance to routing and logistics make it a natural and challenging testbed for evolutionary computation. This report presents our problem formulation, the evolutionary approach and operators we developed, and an experimental analysis of its solution quality and convergence behaviour.

2 Problem

3 Algorithm

4 Experimental part

5 Results and Analysis

6 Conclusions

References